

Mathematics for Social Scientists

Part 3: Linear Algebra III - Matrices and Linear Maps

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Outline

Matrix calculus and linear maps

Matrix Inverse

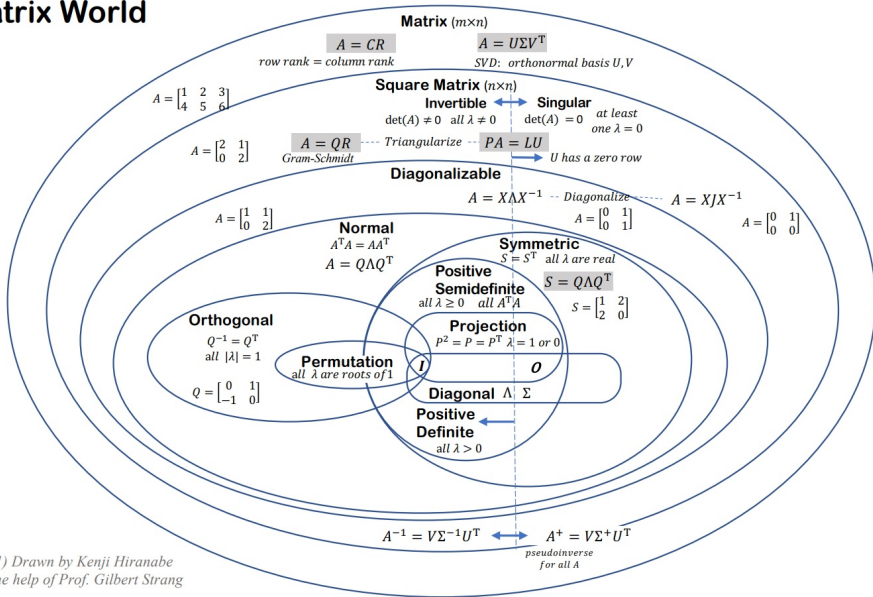
Matrix Rank

Determinant

Linear mappings and matrices

Eigenvalues and Eigenvectors

Matrix World



(v1.4.1) Drawn by Kenji Hiranabe
with the help of Prof. Gilbert Strang

The inverse of a matrix

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix.

Definition 1.

A matrix $B \in \mathbb{R}^{n \times n}$ satisfying

$$AB = BA = I_n$$

is called the **inverse (matrix)** of A . Notation: $B = A^{-1}$.

- The notion of an inverse is only defined for square matrices.
- The inverse of a matrix does not have to exist.
- If it exists, the inverse is uniquely determined. In this case, A is called **invertible** or **non-singular** (opposite: **singular**).

Two questions

Example.

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}. \text{ The inverse is given by } A^{-1} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

Two questions

- How can we determine whether a matrix A is invertible?
- How can we compute the inverse?

Let us first consider the second question!

Computation of the inverse matrix

Consider the invertible matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$.

Form the following **augmented matrix (Gauss tableau)** where the right-hand side is the identity:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 3 & 4 & 1 & 0 & 0 & 1 \end{array} \right).$$

- Reduce the left block to I_3 using elementary row operations **and apply the same operations** to the right block.

Computing A^{-1} step by step (Gauss-Jordan)

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix}, \quad \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 3 & 4 & 1 & 0 & 0 & 1 \end{array} \right).$$

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & -2 & 1 & -3 & 0 & 1 \end{array} \right)$$

$$R_3 \leftarrow R_3 - 2R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

$$R_2 \leftarrow -R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

$$R_1 \leftarrow R_1 - 2R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

Computation of the inverse matrix ctd.

In our specific example, this yields:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right).$$

- Now the inverse is the matrix on the right-hand side of the Gauss tableau.

That is, the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix}$ is

$$A^{-1} = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -1 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$$

Using Inverse to solve the SLE

In order to solve the SLE $Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ one can proceed as follows:

$$\begin{aligned} Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} &\Leftrightarrow A^{-1}Ax = A^{-1} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \Leftrightarrow x = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ &\Leftrightarrow x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Rules of Matrix Inversion

Rules of Matrix Inversion

Let $\mathbf{A}$ and $\mathbf{B}$ be invertible $n \times n$ matrices. Then:

- $\mathbf{A}^{-1}$ is invertible, and $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- $\mathbf{AB}$ is invertible, and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- The transpose $\mathbf{A}'$ is invertible, and $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$
- $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$ whenever $c \neq 0$
- $\det(\mathbf{A}^{-1}) = (\det(\mathbf{A}))^{-1}$

Rank of a matrix

Definition 2.

The **rank of a matrix** $A \in \mathbb{R}^{r \times n}$ is the rank of the SLE $Ax = b$ (formerly written as $(A|b)$) for some $b \in \mathbb{R}^r$.

Notation: $\text{rank}(A) = 2$.

Remark.

This does not depend on the choice of b .

Example.

The rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{pmatrix}$ is equal to 2.

Alternative characterisation via **linear independence!**

Linear independence

Let's imagine you have a collection of vectors. Can you write one of them as a linear combination (addition / scalar multiplication) of others?

If no, then they are *linearly independent*.

If yes, then they are *linearly dependent*.

Example.

- The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent.
- The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are linearly dependent.
- The vectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ are linearly dependent.

Linear Independence ctd

Definition 3 (Linear Independence).

A finite family $v_1, \dots, v_r$ of vectors from $\mathbb{R}^n$ is called **linearly independent** if:

$$\lambda_1 v_1 + \dots + \lambda_r v_r = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_r = 0$$

Otherwise the family is called **linearly dependent**.

Remark.

- A single vector $v \in \mathbb{R}^n$ is linearly independent if and only if $v \neq \mathbf{0}$,
- The vector $\mathbf{0}$ is linearly dependent on any other vector.
- Any family containing the same vector twice is linearly dependent.
- If $r \geq 2$, the vectors $v_1, \dots, v_r$ are linearly dependent, if one v_i is the linear combination of the other vectors.

Linear dependence and rank

Proposition 1.

The rank of a matrix $A^{r \times n}$ is the **maximal** number of linear independent row vectors (considered as elements of $\mathbb{R}^n$).

Example.

$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 4 \\ -2 & -3 & 2 \end{pmatrix}$ has maximal two linearly independent row vectors. The first two rows are linearly independent, whereas all three are not. This is because, third row can be written as a linear combination of the first two $((-2) \cdot r_1 + r_2)$. Therefore, it's rank is 2.

Proposition 2.

Let $A \in \mathbb{R}^{r \times n}$. Then

$$\text{rank}(A) = \text{rank}(A^T).$$

Rank and existence of an inverse

Now: **When does the inverse of a matrix A exist?**

Theorem 3.

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if

$$\text{rank}(A) = n.$$

Corollary 4.

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the SLE

$$Ax = b$$

has a unique solution for every $b \in \mathbb{R}^n$.

Easy criterion to determine whether or not a matrix is invertible: **determinant!**

The determinant of a matrix

Notation: for a matrix $A \in \mathbb{R}^{n \times n}$ we write $A = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$,

where $\mathbf{a}_i$ is the i -th column vector.

Definition 4.

A **determinant** (of order n) is a function

$$\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}, \quad A \mapsto \det(A),$$

such that the following properties are satisfied:

i. For all $s, t \in \mathbb{R}$ and all $i = 1, \dots, n$,

$$\begin{aligned} \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, s\mathbf{a}_i + t\mathbf{a}'_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) \\ = s \cdot \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, \mathbf{a}_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) + t \cdot \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, \mathbf{a}'_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) \end{aligned}$$

ii. $\det(A) = 0$ if $\mathbf{a}_i = \mathbf{a}_j$ for some $i \neq j$,

iii. $\det(I_n) = 1$.

Determinants ctd

Theorem 5.

For all $n \geq 1$, there exists exactly one determinant of order n .

One can explicitly define the determinant for arbitrary n .

On the following slides, we will only give formulae for the cases $n = 2$ and $n = 3$.

Theorem 6.

A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(A) \neq 0$.

The determinant of order 2

Proposition 7.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The determinant of order two is given by the formula

$$\det(A) = ad - bc.$$

Example.

$$\det \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = -2.$$

A closed formula for the inverse of a 2×2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be invertible, that is, $\det(A) = ad - bc \neq 0$. Then, the inverse A^{-1} is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Exercise: Check whether the following matrices are invertible and if yes, compute the inverse:

- $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$
- $B = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$
- $C = \begin{pmatrix} 3 & 3 \\ 1 & 1 \end{pmatrix}$

The determinant of order 3

Proposition 8.

Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$. Then

$$\det(A) = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} - a_{31}a_{22}a_{13}.$$

Example.

$$A = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}. \quad \det(A) = ?$$

The determinant of order 3 (Sarrus' rule)

Example.

$$\begin{aligned}\det(A) &= \det \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} \begin{array}{c} 1 \quad 4 \\ 2 \quad 5 \\ 3 \quad 6 \end{array} \\ &= (1 \cdot 5 \cdot 9 + 4 \cdot 8 \cdot 3 + 7 \cdot 2 \cdot 6) - (3 \cdot 5 \cdot 7 + 1 \cdot 8 \cdot 6 + 4 \cdot 2 \cdot 9) \\ &= (45 + 96 + 84) - (105 + 48 + 72) = 225 - 225 = \boxed{0}.\end{aligned}$$

Some rules for the computation of determinants

Let $A, B \in \mathbb{R}^{n \times n}$.

- Let $A = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$ be diagonal. Then $\det(A) = a_{11} \cdot a_{22} \cdot \dots \cdot a_{nn}$,
- $\text{rank}(A) < n \Leftrightarrow \det(A) = 0$.
- $\det(A \cdot B) = \det(A) \cdot \det(B)$.
- In particular, in the case that A is invertible, $\det(A^{-1}) = (\det(A))^{-1}$.
- $\det(A) = \det(A^\top)$.

Linear mappings and matrices

Definition 5 (Linear map / linear transformation).

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

- $T(x + y) = T(x) + T(y)$ for all $x, y \in \mathbb{R}^n$,
- $T(\lambda x) = \lambda T(x)$ for all $\lambda \in \mathbb{R}, x \in \mathbb{R}^n$.

Matrix representation

For any matrix $A \in \mathbb{R}^{m \times n}$ we can define a linear map

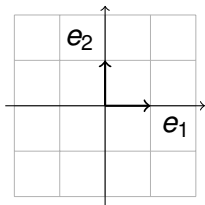
$$T_A(x) := Ax \quad (x \in \mathbb{R}^n),$$

where x is a **vector** in $\mathbb{R}^n$ represented as a column of its coordinates.

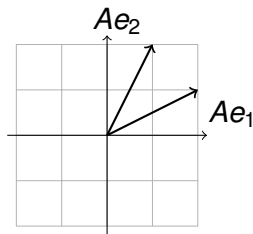
- The matrix A can be seen as a machine that moves every vector x in space to a new vector Ax .
- The transformation is linear: it preserves the origin, scales, and adds directions consistently.
- *Any transformation that can be accomplished with a matrix multiplication is linear.*

Geometric view: what a linear transformation does

Original grid and basis vectors



After applying $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$



- A sends every vector x to Ax .
- The origin remains fixed, and straight lines through the origin stay straight.
- The whole plane is stretched, sheared, or rotated according to what A does to the basis vectors e_1, e_2 .

Some connections between the things we have learned up to now

Let $A \in \mathbb{R}^{n \times n}$. Then:

A has full rank n

$$\Leftrightarrow \det(A) \neq 0$$

$$\Leftrightarrow A \text{ is invertible}$$

$$\Leftrightarrow Ax = b \text{ has a unique solution for every } b \in \mathbb{R}^n$$

$$\Leftrightarrow T_A(x) = Ax \text{ is bijective}$$

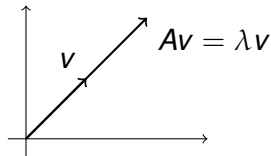
$$\Leftrightarrow \text{columns (and rows) of } A \text{ are linearly independent.}$$

From special directions to eigenvectors and eigenvalues

- For most v , Av points in a different direction than v .
- But if Av is parallel to v , then v and Av are on the same line through the origin.

$$Av = \lambda v$$

- v is an **eigenvector**: its direction is preserved.
- λ is the **eigenvalue**: it tells how v is stretched or shrunk.
- $\lambda > 1$: stretched, $0 < \lambda < 1$: shortened, $\lambda < 0$: flipped.



$$Av = \lambda v \quad \Leftrightarrow \quad (A - \lambda I_n)v = 0$$

This is a homogeneous system with non-trivial solution $v \neq 0$.

Eigenvalues and eigenvectors

Definition 6.

Let $A \in \mathbb{R}^{n \times n}$.

- $\lambda \in \mathbb{R}$ is called **eigenvalue** of A if there exists $\mathbf{0} \neq v \in \mathbb{R}^n$ such that

$$A \cdot v = \lambda \cdot v.$$

- Every $v \neq \mathbf{0}$ satisfying $A \cdot v = \lambda \cdot v$ is called **eigenvector** (for the eigenvalue λ).

Eigenvalues and Eigenvectors ctd

Example.

The matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$ with the corresponding examples of eigenvectors:

$$x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Exercise: Check this!

Solution: eigenvalues

For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

we compute the **characteristic polynomial**:

$$P_A(\lambda) = \det(A - \lambda I_2) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix}.$$

We expand the determinant:

$$P_A(\lambda) = (2 - \lambda)(2 - \lambda) - 1 = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Factor:

$$P_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3),$$

so the eigenvalues are

$$\lambda_1 = 1 \quad \text{and} \quad \lambda_2 = 3.$$

Solution: eigenvectors for $\lambda_1 = 1$

We solve

$$(A - I_2)v = 0 \quad \text{for} \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

$$A - I_2 = \begin{pmatrix} 2-1 & 1 \\ 1 & 2-1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

We want

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which gives one equation:

$$v_1 + v_2 = 0 \quad \Rightarrow \quad v_1 = -v_2.$$

A non-zero solution of this system is obtained by taking $v_2 = 1$, which gives $v_1 = -1$:

$$v = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

An eigenvector for $\lambda_1 = 1$ is therefore

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Any non-zero scalar multiple of this vector is also an eigenvector — we care about the direction, not the length.

Solution: eigenvectors for $\lambda_2 = 3$

Now solve

$$(A - 3I_2)v = 0 \quad \text{for} \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

$$A - 3I_2 = \begin{pmatrix} 2-3 & 1 \\ 1 & 2-3 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

We want

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which gives

$$-v_1 + v_2 = 0 \quad \Rightarrow \quad v_1 = v_2.$$

Choose e.g. $v_1 = 1$, then $v_2 = 1$:

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

An eigenvector for $\lambda_2 = 3$ is therefore

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Any non-zero scalar multiple of this vector is also an eigenvector.

So:

$\lambda_1 = 1$ with direction $(-1, 1)$,

$\lambda_2 = 3$ with direction $(1, 1)$.

Getting the eigenvalues

Q: How to determine eigenvalues and eigenvectors?

Proposition 9.

For $A \in \mathbb{R}^{n \times n}$ the following statements are equivalent:

- λ is an eigenvalue of A .
- $\det(A - \lambda \cdot I_n) = 0$.

Thus, the search for eigenvalues can be reduced to the question of finding the roots/zeros of the so-called *characteristic polynomial*:

$$P_A(\lambda) := \det(A - \lambda \cdot I_n).$$

Remark.

$P_A(\lambda)$ is a polynomial of degree n .

Examples

- $A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix} \rightsquigarrow P_A(\lambda) = \lambda^2 - 3\lambda + 2.$

- $B = \begin{pmatrix} 0 & -1 & 1 \\ -3 & -2 & 3 \\ -2 & 2 & 3 \end{pmatrix} \rightsquigarrow P_B(\lambda) = -\lambda^3 + \lambda^2 + 13\lambda - 13.$

Getting the eigenvectors

- Recall that the eigenvector equation

$$Ax = \lambda x$$

can be written as

$$(A - \lambda I_n)x = \mathbf{0}.$$

- This is a **homogeneous system of linear equations**.
- We are looking for *non-zero* solutions $x \neq 0$.
- These special solutions indicate directions that A does not rotate, only stretches or shrinks.

Geometric meaning

Each eigenvalue λ corresponds to a line (or plane) through the origin along which all vectors keep the same direction under the transformation.

Getting the eigenvectors ctd

Example (see example from before).

$A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$.

For $\lambda_1 = 1$ we form the system

$$(A - I_2)x = 0 \implies \begin{cases} -2x_1 + 6x_2 = 0, \\ -x_1 + 3x_2 = 0. \end{cases}$$

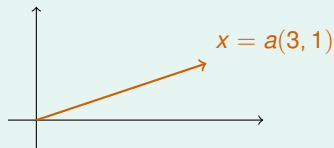
Both equations describe the same line:

$$x_1 = 3x_2.$$

Thus, every vector on that line

$$x = a \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad a \neq 0,$$

is an eigenvector for $\lambda = 1$.



Question: What is the direction for $\lambda_2 = 2$?